



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No : 12

TITLE : Write a Java program to create and use user-defined packages.

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.



Code and Output Screenshots

-> Class inside the package mypack

```
mypack > J Myclass.java > Java > Myclass
1  package mypack;
2
3  public class Myclass {
4      public void display() {
5          System.out.println("This is a package example.");
6          System.out.println("This code was written by Atharva.");
7      }
8  }
```

-> Importing class in main program

```
J Main11.java > ...
1  import mypack.Myclass;
2
3  public class Main11 {
4      Run main | Debug main | Run | Debug
5      public static void main(String[] args) {
6          Myclass obj = new Myclass();
7          obj.display();
8      }
9  }
```

PROBLEMS 21

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Main11.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Main11
This is a package example.
This code was written by Atharva.
PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



Conclusion

This experiment demonstrated the implementation and use of packages in Java. A user-defined package was created, imported, and accessed successfully from another class. The program executed correctly, showing how packages help organize classes, improve code reusability, and avoid naming conflicts. Overall, the experiment enhanced understanding of Java package creation, importing, compilation, and execution.

Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

Ans)

- Class for student details in “details” package

```
Details > J stdtls.java > ...
1  package details;
2
3  public class Stdtls {
4      String name;
5      int age;
6      String branch;
7
8      public Stdtls(String name, int age, String branch) {
9          this.name = name;
10         this.age = age;
11         this.branch = branch;
12     }
13
14     public void display() {
15         System.out.println("Name: " + name);
16         System.out.println("Age: " + age);
17         System.out.println("Branch: " + branch);
18     }
19 }
20
```



- Class for teacher details in “details” package

```
details > J Tdtls.java > ...
1  package details;
2
3  public class Tdtls {
4      String name;
5      int tid;
6      String dept;
7
8      public Tdtls(String name, int tid, String dept) {
9          this.name = name;
10         this.tid = tid;
11         this.dept = dept;
12     }
13
14     public void display() {
15         System.out.println("Name: " + name);
16         System.out.println("Teacher ID: " + tid);
17         System.out.println("Department: " + dept);
18     }
19 }
20
```



- Package import and object creation

```
J Main12.java > ...
1  import details.stdtls;
2  import details.Tdtls;
3  public class Main12 {
    Run main | Debug main | Run | Debug
4      public static void main(String args []){
5          stdtls student1 = new stdtls("Atharva", 18, "CSE");
6          stdtls student2 = new stdtls("Rachit", 19, "ENTC");
7
8          Tdtls t1 = new Tdtls("Dr Ketan Kotecha", 10, "AI and ML");
9          Tdtls t2 = new Tdtls("Dr Rupali Nagar", 55, "Physics");
10
11         System.out.println("Student 1 Details:");
12         student1.display();
13
14         System.out.println("\nStudent 2 Details:");
15         student2.display();
16
17         System.out.println("\nTeacher 1 Details:");
18         t1.display();
19
20         System.out.println("\nTeacher 2 Details:");
21         t2.display();
22     };
23 }
24
```



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- Output

```
PROBLEMS 35 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main12.java && java Main12
Student 1 Details:
Name: Atharva
Age: 18
Branch: CSE

Student 2 Details:
Name: Rachit
Age: 19
Branch: ENTC

Teacher 1 Details:
Name: Dr Ketan Kotecha
Teacher ID: 10
Department: AI and ML

Teacher 2 Details:
Name: Dr Rupali Nagar
Teacher ID: 55
Department: Physics

[Done] exited with code=0 in 0.522 seconds
|
```



2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

Ans)

- Class Bdtls in package library to display book details

```
library > J Bdtls.java > Java > Bdtls
1  package library;
2
3  public class Bdtls {
4      int Bid;
5      String author;
6      int price;
7
8      public Bdtls(int Bid, String author, int price) {
9          this.Bid = Bid;
10         this.author = author;
11         this.price = price;
12     }
13
14     public void display() {
15         System.out.println("Book ID: " + Bid);
16         System.out.println("Author: " + author);
17         System.out.println("Price: " + price);
18     }
19 }
20
```



- Importing the package, object creation and output.

```
J Main13.java > ...
1  import library.Bdtls;
2  public class Main13 {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4          Bdtls book1 = new Bdtls(101, "J.K. Rowling", 600);
5          Bdtls book2 = new Bdtls(150, "E Balaguruswamy", 400);
6          Bdtls book3 = new Bdtls(110, "Homer", 1000);
7
8          System.out.println("Book 1 Details:");
9          book1.display();
10
11         System.out.println("\nBook 2 Details:");
12         book2.display();
13
14         System.out.println("\nBook 3 Details:");
15         book3.display();
16
17     }
18 }
19
```

PROBLEMS 35 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main13.java && java Main13
Book 1 Details:
Book ID: 101
Author: J.K. Rowling
Price: 600

Book 2 Details:
Book ID: 150
Author: E Balaguruswamy
Price: 400

Book 3 Details:
Book ID: 110
Author: Homer
Price: 1000

[Done] exited with code=0 in 0.511 seconds
```



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